

GULKANA, AK, USA

WMO: 702710

Lat: 62.159N

Lon: 145.459W

Elev: 1562

StdP: 13.89

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26425

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.1	(c) -32.3	(d) -34.5	(e) 0.8	(f) -28.3	(g) -32.0	(h) 1.0	(i) -26.1	(j) 26.2	(k) 32.7	(l) 23.8	(m) 33.5	(n) 1.2	(o) 310	(p) 0.758

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	78.2	58.3	74.5	56.8	70.9	55.3	60.2	74.7	58.3	71.5	56.5	68.2	6.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.1	65.9	62.5	52.3	61.6	61.2	50.6	57.8	59.9	27.1	75.2	25.8	71.4	24.6	67.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.5	20.5	17.9	-40.4	82.3	4.0	3.9	-43.3	85.1	-45.6	87.4	-47.9	89.5	-50.7	92.3		
			-33.3	62.1	2.9	2.3	-35.4	63.8	-37.1	65.1	-38.7	66.4	-40.8	68.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.9	-1.2	6.1	15.0	32.8	45.9	55.0	57.9	53.9	44.1	28.6	7.0	0.1
	DBStd	24.40	17.36	14.67	12.82	8.33	6.21	5.44	4.74	5.18	6.43	10.88	15.32	16.40
	HDD50	8296	1586	1228	1085	516	158	12	2	19	190	664	1290	1546
	HDD65	13186	2051	1648	1550	966	591	303	225	346	627	1128	1740	2011
	CDD50	595	0	0	0	0	32	163	247	139	13	1	0	0
	CDD65	10	0	0	0	0	0	4	5	1	0	0	0	0
	CDH74	360	0	0	0	0	10	126	146	78	0	0	0	0
	CDH80	52	0	0	0	0	0	26	18	8	0	0	0	0
	WSAvg	5.5	3.1	3.7	5.6	6.9	7.5	7.2	7.0	6.6	6.7	5.0	3.6	3.1
	PrecAvg	11.0	0.5	0.5	0.3	0.2	0.7	1.6	1.8	1.6	1.5	0.9	0.6	0.8
(15) Precipitation	PrecMax	14.9	1.5	1.7	1.3	0.8	2.2	4.1	3.3	3.9	3.9	2.4	2.9	2.6
	PrecMin	5.3	0.0	0.0	0.0	0.0	0.0	0.3	0.5	0.0	0.1	0.1	0.0	0.0
	PrecStd	2.5	0.3	0.4	0.3	0.2	0.6	0.8	0.7	0.8	0.9	0.6	0.6	0.6
	PrecStd	2.5	0.3	0.4	0.3	0.2	0.6	0.8	0.7	0.8	0.9	0.6	0.6	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	39.9	46.8	62.0	75.5	84.4	82.2	80.9	66.2	54.3	40.1	39.1
		MCWB	36.4	34.5	38.1	46.4	53.7	61.2	61.3	60.1	51.7	47.4	35.1	35.8
	2%	DB	36.4	36.6	40.8	54.5	70.1	77.4	78.0	76.0	62.0	48.8	37.6	35.8
		MCWB	33.8	32.5	34.0	42.1	50.8	56.9	59.0	58.1	49.7	41.8	34.0	33.0
	5%	DB	31.5	33.1	37.6	50.2	65.4	73.0	74.4	70.8	58.7	45.7	34.5	31.8
		MCWB	28.7	29.9	31.7	39.0	48.2	55.0	57.9	56.2	48.4	39.7	32.1	29.6
	10%	DB	24.5	28.5	34.4	46.9	60.6	68.9	70.8	66.3	55.9	42.9	29.4	25.0
		MCWB	22.9	25.8	29.3	37.0	46.3	53.6	56.5	54.0	47.1	38.1	27.7	23.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.0	35.4	38.2	46.9	54.2	62.5	62.9	61.9	53.0	47.9	36.3	36.5
		MCDB	39.5	38.1	45.8	60.9	72.7	80.0	78.7	77.3	61.3	53.6	38.3	38.3
	2%	WB	34.1	32.9	34.7	42.8	51.5	58.6	60.5	59.3	51.0	43.0	34.2	33.2
		MCDB	36.8	36.2	39.9	53.8	68.2	74.4	74.4	73.1	59.3	47.1	37.0	35.4
	5%	WB	29.0	30.1	32.2	39.5	49.2	56.2	58.9	57.0	49.6	40.4	32.0	29.6
		MCDB	31.6	33.3	37.3	48.8	62.8	70.0	72.0	68.8	57.4	44.8	34.7	31.8
	10%	WB	23.6	26.0	29.7	37.3	47.2	54.3	57.2	55.0	48.0	38.3	27.8	23.9
		MCDB	25.4	28.9	34.5	46.1	59.6	66.8	68.9	64.5	54.7	42.7	29.5	25.8
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	19.6	25.1	22.2	22.7	22.6	20.7	21.1	19.7	16.2	15.1	14.9
		MCDBR	16.1	18.1	21.6	26.6	32.5	30.3	29.1	30.5	25.2	16.5	14.5	17.2
		MCWBR	14.1	15.2	15.8	15.8	16.0	13.8	13.3	15.1	14.8	11.2	12.4	15.4
	5% WB	MCDBR	15.7	18.0	19.5	25.3	28.5	27.1	25.9	26.6	22.2	15.8	13.9	16.3
		MCWBR	13.8	15.4	14.3	15.2	14.5	13.0	12.2	13.6	13.5	11.3	12.1	14.8
(40) Clear-Sky Solar Irradiance	taub		0.207	0.263	0.272	0.356	0.356	0.368	0.363	0.358	0.303	0.279	0.215	0.188
	taud		2.050	2.184	2.246	2.094	2.342	2.361	2.402	2.420	2.556	2.449	2.057	1.908
	Ebn at Noon		156	223	268	260	273	271	270	261	260	222	145	92
	Edh at Noon		11	20	28	40	34	35	32	29	21	16	10	7
(44) All-Sky Solar Radiation	RadAvg		81	318	875	1459	1722	1728	1560	1214	783	358	118	40
	RadStd		9	34	69	80	96	132	103	106	81	43	17	3

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+4
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page